

Facing Scalable Sequence Data: Theory, Algorithms, and Applications

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[It is just a framework of research statement]

1 Background

The era of large-scale biological sequence data has arrived, with collections now reaching the petabyte scale and growing rapidly, fundamentally reshaping how genomic analysis is performed. Storage, searching, querying, comparison, and many types of large-scale analysis all face unprecedented computational challenges. k -mer-based and sketching methods have emerged as powerful approaches for handling massive sequence collections due to their scalability and efficiency. However, several fundamental gaps remain.

First, repetitive sequences pose a major challenge for k -mer-based and sketching methods. Repeats violate core statistical assumptions, leading to a lack of rigorous theory and often degraded performance in practice.

Second, many realistic biological scenarios remain insufficiently modeled, such as read-mapping settings and evolutionary processes involving branching mutations (e.g., two-child models). These scenarios currently lack solid theoretical foundations and efficient algorithms.

Third, there is still a shortage of specialized k -mer-based and sketching algorithms tailored to important biological problems, including ANI estimation and the analysis of highly repetitive regions such as centromeres.

2 Current Research

My current research addresses these challenges by building a unified framework that connects statistical theory, scalable algorithms, and practical genomic applications.

1. I developed a repeat-aware theoretical framework for modeling mutations in repetitive sequences using k -mer statistics, leading to more accurate estimators and a family of methods designed for different data scenarios.

2. I applied these repeat-aware estimators to centromere analysis, estimating the T_i/T_v ratio and building a k -mer-based metric for measuring distances between centromeres.
3. I introduced a theoretical framework for read-mapping-like settings, enabling mutation-rate estimation when comparing reads to whole genomes.
4. Based on these models, I designed fast and accurate approaches for ANI estimation, providing a better trade-off between computational efficiency and accuracy.
5. I developed a theoretical framework for two-child mutation models, enabling inference of generation mutation rates from descendant sequences.
6. I established a statistical framework for k -mer sharing patterns, allowing confidence scoring of search results and improving the reliability of large-scale query systems such as Logan.

3 Future Vision

Looking forward, my research program aims to advance scalable and trustworthy alignment-free genomics along three complementary directions.

1. **Theory.** I will continue to develop theoretical models that better reflect realistic biological scenarios, including extensions that incorporate indels and mutation-rate estimators based on read sets.
2. **Algorithms.** I will design new k -mer-based and sketching algorithms tailored to additional biological problems, such as metagenomic profiling, binning, and large-scale sequence search.
3. **Applications.** I aim to contribute theoretical and algorithmic foundations to large collaborative efforts, including Logan, the 1000 Human Pangenome Project, and centromere genomics studies, enabling scalable and interpretable analysis of complex genomic data.

References